

## Llista 1. Estadística Descriptiva. Solucions

### Exercici 1

| Fills      | Freq. abs | Freq. abs. acum. | Freq. rel.  | Freq. rel. acum. |
|------------|-----------|------------------|-------------|------------------|
| <b>0</b>   | 52        | 52               | 0.26        | <b>0.26</b>      |
| <b>1</b>   | 48        | 100              | <b>0.24</b> | 0.5              |
| <b>2</b>   | <b>45</b> | 145              | 0.225       | 0.725            |
| <b>3</b>   | 25        | <b>170</b>       | 0.125       | 0.85             |
| <b>4</b>   | 28        | 198              | 0.14        | <b>0.99</b>      |
| <b>5++</b> | <b>2</b>  | 200              | 0.1         | 1                |

### Exercici 2

| Notes     | Freq. abs | Freq. abs. acum. | Freq. rel. | Freq. rel. acum. |
|-----------|-----------|------------------|------------|------------------|
| <b>0</b>  | 4         | 4                | 0.1        | 0.1              |
| <b>1</b>  | 4         | 8                | 0.1        | 0.2              |
| <b>2</b>  | 5         | 13               | 0.125      | 0.325            |
| <b>3</b>  | 4         | 17               | 0.1        | 0.425            |
| <b>4</b>  | 6         | 23               | 0.15       | 0.575            |
| <b>5</b>  | 6         | 29               | 0.15       | 0.725            |
| <b>6</b>  | 3         | 32               | 0.075      | 0.8              |
| <b>7</b>  | 3         | 35               | 0.075      | 0.875            |
| <b>8</b>  | 3         | 38               | 0.075      | 0.95             |
| <b>9</b>  | 1         | 39               | 0.025      | 0.975            |
| <b>10</b> | 1         | 40               | 0.025      | 1                |

```
> x <- c(7, 3, 2, 4, 5, 1, 8, 6, 1, 5, 3, 2, 4, 9, 8, 1, 0, 2, 4, 1, 2, 5, 6, 5, 4, 7, 1, 3,
```

```
## b)
```

```
> mean(x)
```

```
[1] 4.075
```

```
> sd(x)*(length(x)-1)/length(x)      # Desviació típica
```

```
[1] 2.601802
```

```
> median(x)
```

```
[1] 4
```

Moda = 4 i 5.

### Exercici 3

| Interval | Marca      | Freq.<br>abs | Freq. abs.<br>acum. | Freq. rel. | Freq. rel.<br>acum. |
|----------|------------|--------------|---------------------|------------|---------------------|
| [0,5)    | <b>2.5</b> | 10           | 10                  | 0.32       | 0.32                |
| [5,7)    | <b>6</b>   | 8            | 18                  | 0.26       | 0.58                |
| [7,10)   | <b>8.5</b> | 9            | 27                  | 0.29       | 0.87                |
| [10,12]  | <b>11</b>  | 4            | 31                  | 0.13       | 1                   |

a) Q1  $\rightarrow$  [0,5) Q2  $\rightarrow$  [5,7) Q3  $\rightarrow$  [7,10)

b)  $Me = L_i + (L_{i+1} - L_i) \frac{\frac{n}{2} - N_{i-1}}{n_i} = 5 + (7 - 5) \frac{\frac{31}{2} - 10}{8} = 6.375$

c) 20%-percentil  $\rightarrow$  [0,5) 85%-percentil  $\rightarrow$  [7,10)

#### Exercici 4

```
> stol <- c(271,428,381,366,411,193,178,178,427,180)
```

```
## a)
```

```
> sort(stol)
[1] 178 178 180 193 271 366 381 411 427 428
> median(stol)
[1] 318.5
> table(stol)
stol
178 180 193 271 366 381 411 427 428
  2   1   1   1   1   1   1   1   1
```

```
## b)
```

```
> var(stol); sd(stol)
[1] 12465.79
[1] 111.6503
```

c)  $y_i = kx_i$

Mitjana:

$$\bar{y} = \sum_{i=1}^n y_i = \sum_{i=1}^n kx_i = k \sum_{i=1}^n x_i = k\bar{x}$$

Variància:

$$s_y^2 = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (kx_i - k\bar{x})^2 = k^2 \sum_{i=1}^n (x_i - \bar{x})^2 = k^2 s_x^2$$

```
> k <- 3.28084
> mean(stol)*k
[1] 988.5171
> var(stol)*k^2
[1] 134180.6
```

### Exercici 5

```
## a)
> # Rang
> max(temperatura)-min(temperatura)
[1] 26.5
> # Rang interquantil.lic
> summary(temperatura)[5]-summary(temperatura)[2]
3rd Qu.
6.875
```

b)

c) Número de classes: 7

Longitud dels intervals: 4 °C

d)

| Temper.<br>(Marca) | Freq. abs | Freq. abs. acum. | Freq. rel. | Freq. rel. acum. |
|--------------------|-----------|------------------|------------|------------------|
| [65,69)            | 1         | 1                | 0.033      | 0.033            |
| [69,73)            | 5         | 6                | 0.167      | 0.2              |
| [73,77)            | 4         | 10               | 0.133      | 0.333            |
| [77,81)            | 7         | 17               | 0.233      | 0.566            |
| [81,85)            | 8         | 25               | 0.267      | 0.833            |
| [85,89)            | 3         | 28               | 0.1        | 0.933            |
| [89,93]            | 2         | 30               | 0.067      | 1                |

```
## e)
> (7/30)*3/4
[1] 0.175 # 17.5%
```

### Exercici 6

|                 | N   | SN  | MSN |      |
|-----------------|-----|-----|-----|------|
| <b>Impur</b>    | 19  | 29  | 24  | 72   |
| <b>No impur</b> | 560 | 269 | 497 | 1326 |
|                 | 579 | 298 | 521 | 1398 |

a)  $72/1398 = 0.0515 = 5.15\%$

b)  $579/1398 = 0.4142 = 41.42\%$

c)  $497/1398 = 0.3555 = 35.55\%$

d)  $(29+24)/1398 = 0.0379 = 3.79\%$

### Exercici 7

```
> eritrocits <- c(4.2, 5.7, 6.1, 3.8, 4.5, 5.2, 4.6, 4.3)
> ferro <- c(33, 48, 53, 44, 41, 39, 42, 36)

## a)
> mean(eritrocits); mean(ferro)
[1] 4.8
[1] 42

## b)
> var(eritrocits); var(ferro)
[1] 0.6285714
[1] 41.14286
> sd(eritrocits); sd(ferro)
[1] 0.792825
[1] 6.41427

## c)
> # Coeficient de variació
> coef_var(eritrocits); coef_var(ferro)
[1] 0.1651719
[1] 0.1527207

> ## d)
> cov(eritrocits,ferro)
[1] 3.6
> cor(eritrocits,ferro)
[1] 0.7079099
```

### Exercici 8

```
a) 16 anys - Enunciat: dim(previsions)
b) Enunciat: mean(...), var(...), cov(...)

## c)
cor(Demanda, Vendes)
[1] 0.8194537
# Correlació lineal moderada positiva

## d)
> reg

Call:
```

```
lm(formula = Vendes ~ Demanda)

Coefficients:
(Intercept)      Demanda
      1.43558      0.02354
# També es pot trobar a l'enunciat: summary(reg) - Coefficients

## e)
> new.x <- data.frame(Demanda = c(150, 210, 380))
> predict(reg, newdata = new.x)
      1      2      3
4.966077 6.378275 10.379502

## f)
# Valors ajustats
> v_ajustats <- predict(reg)
> v_ajustats[1:3]
      1      2      3
6.142909 6.613641 10.850235
# Residus
> residus <- Vendes-v_ajustats
> residus[1:3]
      1      2      3
2.8570913 -0.6136413  1.1497654
```

### Exercici 9

a) Enunciat: mean(...), var(...), cov(...)

Unitats de mesura: Mitjanes g i kcal, variàncies g<sup>2</sup> i kcal<sup>2</sup>, covariància g · kcal.

```
## b)
cor(carbohidrats, calories)
[1] 0.7247129
# Correlació lineal moderada positiva
```

```
## c)
```

```
> reg
```

Call:

```
lm(formula = calories ~ carbohidrats)
```

Coefficients:

```
(Intercept)  carbohidrats
      93.00      10.65
```

Coefficient de determinació: 0.4659 → 46.6% - Enunciat: Adjusted R - squared.

```
## d)
> new.x <- data.frame(carbohidrats = 5)
> predict(reg, newdata = new.x)
      1
146.2287
```

### Exercici 10

```
> pares <- c(165, 160, 170, 162, 173, 157, 178, 168, 173, 170)
> fills <- c(173, 168, 173, 167, 175, 168, 173, 170, 174, 170)
```

```
> ## a)
> mean(pares); mean(fills)
[1] 167.6
[1] 171.1
> var(pares); var(fills)
[1] 42.93333
[1] 8.1
> sd(pares); sd(fills)
[1] 6.552353
[1] 2.84605
```

```
## b)
# Covariancia
> cov(pares, fills)
[1] 14.71111
# Coeficient de correlacio
> cor(pares, fills)
[1] 0.7888704
# Coeficient de determinacio
> cor(pares, fills)^2
[1] 0.6223165
```

```
## c)
> reg <- lm(fills~pares)
> reg
```

Call:

```
lm(formula = fills ~ pares)
```

Coefficients:

| (Intercept) | pares  |
|-------------|--------|
| 113.6718    | 0.3427 |

```
## d)
> new.p <- data.frame(pares = 166)
```

```
> predict(reg, newdata = new.p)
      1
170.5518
```

### Exercici 11

```
> x <- c(1, 2, 6, 3, 5, 10)
> y <- c(3.8, 8.2, 19.1, 9.5, 15, 31.1)
```

## a)

```
> reg <- lm(y~x)
> reg
```

Call:

```
lm(formula = y ~ x)
```

Coefficients:

```
(Intercept)          x
      1.064         2.975
```

## b)

# Covariancia

```
> cov(x, y)
```

```
[1] 31.83
```

# Coeficient de correlacio

```
> cor(x, y)
```

```
[1] 0.9971597
```

# Coeficient de determinacio

```
> cor(x, y)^2
```

```
[1] 0.9943274
```

## c)

# Valors ajustats

```
> y_ajustats <- predict(reg)
```

```
> y_ajustats
```

```
      1      2      3      4      5      6
4.038318 7.013084 18.912150 9.987850 15.937383 30.811215
```

# Residus

```
> residus <- y-y_ajustats
```

```
> residus
```

```
      1      2      3      4      5      6
-0.2383178 1.1869159 0.1878505 -0.4878505 -0.9373832 0.2887850
```